Abstract

The paper "A Bayesian Test for the Hot Hand Phenomenon" by Wetzels et al. introduces a Bayesian test for assessing the presence of the hot hand phenomenon. In sports the hot hand phenomenon posits that the performance of a player is marked by intermittent bursts of remarkable ability. The paper's Bayesian test compares a constant performance model to a hidden Markov model with two states, hot and cold, and a probability of switching between those states. The test is assessed through a simulation study and applied to real data sets from free throw basketball performances. Yet, to comprehensively cover this phenomenon in basketball an analysis of overall throw statistics, not just free throws, seems more appropriate. Considering overall throw statistics, including regular field goals, should allow for a more holistic view and better understanding of streakiness. It highlights the importance of a wider range of performance metrics and their subsequent combination to accurately assess the hot hand phenomenon in basketball.

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Abbreviations

By order of appearance

HHP	Hot Hand Phenomenon
CPM	Constant Performance Model
HMM	Hidden Markov Model
α	Transition Probability
BF	Bayes Factor
NBA	National Basketball Association
OT	Overtime
C	Center
SG	Shooting Guard
SF	Small Forward
FG%	Field Goal Percentage
H0	Null Hypothesis
H1	Alternative Hypothesis

1. Introduction

"The hot hand phenomenon refers to the popular notion that the performance of sports players is punctuated by streaks of exceptional performance. During these streaks, the player is said to be 'hot', or even 'on fire'."(Wetzels et al., 2016, p. 200) These streaks are widely recognized in sports, especially by fans. However, the validity of this loosely defined phenomenon varies across different sports[1] and even more so when accounting for the models used to analyze it. The effectiveness of the chosen model is influenced by numerous factors inherent to each sport, shaping the perception of a true underlying streak.

The article A Bayesian Test for the Hot Hand Phenomenon by Wetzels et al. (2016) presents a novel statistical test for assessing the presence of said phenomenon with binary sequences of successes and failures. The study develops a Bayesian test that compares a constant performance model (CPM) to a hidden Markov model (HMM) with two states, representing either hot and cold performance, and a probability α of switching between states. These models proceed to undergo a Bayes factor (BF) test. The test is assessed through a simulation study and applied to real data sets split in seasonal basketball performance of two players in the National Basketball Association (NBA). The NBA application set to assess streaky performance in sports focuses particularly on basketball free throw shooting. (Wetzels et al., 2016, p. 200)

The articles analysis suggests that assessing clear evidence for or against streakiness is difficult, except for very long data sequences or extreme, evidence backed forms of streakiness. The mixed evidence for streakiness suggests the utilized free throw data may not be informative enough to sufficiently distinguish between a streaky and a non-streaky model. (Wetzels et al., 2016, p. 206)

For the focus of this paper I propose that only assessing free throw shooting rates is insufficient. I argue, as free throws only make up a marginal portion of a typical NBA game their subsequent influence on a player does not encapsulate the full picture of said player's streakiness. The introduced model and process will stay the same to make of a fitting results comparison on how streakiness might change with this overall shot introduction.

For a more in depth analysis I thus apply the same BF test on the entire shooting performance of several games. This should, in theory, recreate a broader picture of streakiness among the analysed players. The research of this paper consequently aims to provide a deeper understanding of practical utilization of a HMM and an inherent BF test, as well as to critically evaluate the results for streakiness in A Bayesian Test for the Hot Hand Phenomenon.

Put to the test, under the new proposal, the analysed player's streakiness was indeed different from the initial results of the BF test by Westels et al. but it was not strong. Yet, these results have to be taken with scrutiny, as only two players were compared over a short time frame. Iterating the same model over other players with roughly equal playing and same positions over the same seasons or games could result in more robust findings.

[1] A streak in baseball, i.e., widely differs from a shooting streak in basketball, which again is completely different from an American football quarterback's scoring streak.

2. Theory and Methodology

In this section I will present the given model and test of the article, as well as shortly mention its second model, a visual discrimination task. I then proceed to integrate my own methodology extension regarding the issue.

2.1 Bayesian Test by Wetzels et al.

I will firstly summarize the article's structure and process of theory behind a BF test for the HHP.[2] Wetzels et al. introduce and discuss several ideas on how to approach the HHP but ultimately settle on structuring their model after Jim Albert (Albert, 1993)(Albert and Williamson, 2001)(Albert, 2008), albeit including a few modifications.

2.1.1 Existing Theory

The authors posit:

> "Our test for the hot hand phenomenon uses the Bayes factor to quantify the adequacy of a constant performance model against that of a streaky performance model. The streaky performance model is a hidden Markov model with two states and one transition probability. [...] We do not assign the transition probability α a fixed value, but rather treat it as a free parameter." (Wetzels et al., 2016, p. 201)

and add:

> "We assume that a player can switch states at any time point t. Hence the binary random variable that indicates success or failure at time t follows a Bernoulli distribution with a success probability that depends on the hidden state at time t. The underlying process is assumed to follow a stationary first-order Markov chain, meaning that the probability of being in a certain state at time t depends only on the state occupied at time $t-1$." (Wetzels et al., 2016, p. 201)

Wetzels et al. go about several steps which are required to define the desired model:

- A two-state Bernoulli HMM is utilized to represent streaky performance, with two states, $S_t = 1$ denoting a *hot* state and $S_t = 0$ a *cold* state.
- The one-step transition probability matrix $\Gamma = (\gamma_{ij})_{i,j\in\{1,2\}}$ contains the probability of switching from the hot to the cold state and vice versa with sticky states being $\alpha < .5$, only those sticky states produce performance that is consistent with streakiness and the HHP.
- The state-dependent sequence of random variables $Y_t : t \in \mathbb{N}$ produces the sequence of observations $y_t, t \in 1, ..., T.Y_t$ is distributed according to a Bernoulli distribution for all t. Here $Y_t = 0$ indicates failure (e.g., a miss) and $Y_t = 1$ indicates success (e.g., a hit).

[2]For a deeper comprehension it is of course more suitable to read the full article itself, as I will only include what is deemed to be necessary for this paper's analysis.

- A player can have success both in the hot and in the cold state. However, the probability of success is by definition higher in the hot than in the cold state.
- The random variable Y_t therefore has a different Bernoulli distribution $Y_t \sim Bern(p_{S_t})$ depending on the current state S_t. The probability of success in the hot state is given by $\theta_h = p(Y_t = 1|S_t = 1)$, and the probability of success in the cold state by $\theta_c = p(Y_t = 1|S_t = 0)$.
- Two diagonal matrices were created for the success and failure probabilities for both states:

$$p(y_t = 1) = \begin{pmatrix} \theta_h & 0 \\ 0 & \theta_c \end{pmatrix}$$

$$p(y_t = 1) = \begin{pmatrix} 1 - \theta_h & 0 \\ 0 & 1 - \theta_c \end{pmatrix}$$

- The CPM is a model that assumes a single, constant success probability $\theta = p(Y_t = 1)$ for all time points $t \in \mathbb{N}$. In the case of the CPM the observed data y_t are the outcome of a sequence of independent random variables $Y_1, Y_2, Y_t, ..., Y_T$ with $Y_t \sim Bern(\theta)$ for all $t \in \mathbb{N}$.

With this pre-work the authors proceed to introduce two likelihood models. The two-state Bernoulli HMM's likelihood is defined as:[3]

$$L_{HMM} = \delta p(y_1)\Gamma p(y_2) \cdots \Gamma p(y_T)\mathbf{1'}$$

> "Where **1'** is a 2-dimensional row vector and δ is the initial distribution of the Markov chain. Here we assume that a player is equally likely to start in one or the other state which means $\delta = (1/2, 1/2)$. Hence, our two-state HMM has three free parameters: the probability θ_h of success in the hot state, the probability θ_c of success in the cold state, and the probability α of switching between states." (Wetzels et al., 2016, p. 201)

And the likelihood LCPM for the CPM as:

$$L_{CPM} = \theta^k(1 - \theta)^{T-k}$$

> "With k being the number of successes in the observed sequence of length T."(Wetzels et al., 2016, p. 202)

The HMM is compared to the CPM using the mentioned BF test. The BF is given by the ratio of marginal likelihoods:[4]

$$BF_{HC} = \frac{p(Y^{(T)} = y^{(T)}|\text{HMM})}{p(Y^{(T)} = y^{(T)}|\text{CPM})}$$

[3]Here the article further refers to: (Zucchini and MacDonald, 2009).

[4]Here the article further refers to: (Jeffreys, 1961)(Kass and Raftery, 1995).

"Where Y(T) and y(T) denote the time-ordered vector of random variables $Y_1, Y_2, ..., Y_T$ and observations $y_1, y_2, ..., y_T$, respectively. The Bayes factor indicates the change from prior to posterior odds brought about by the data. When $BF_{HC} = .20$, for instance, this indicates that the observed data are 5 times more likely to occur under the CPM than under the HMM. The marginal likelihoods are obtained by integrating out the model parameters over the prior distribution."[5] (Wetzels et al., 2016, p. 202)

Results were made interpretable with a discrete classification scheme by Jefferys (Jeffreys, 1961). Where negative results of the BF classify the CPM or non-streakiness and positive results classify several levels of the HMM or streakiness depending on the size of a positive factor number.

BF_{HC}	Strength of evidence
0–1	Negative (supports CPM)
1–3	Not worth more than a bare mention
3–10	Substantial
10–30	Strong
30–100	Very strong
>100	Decisive

Figure 1: Classification table after Jefferys

2.1.2 Simulation Study

In the simulation study, the authors aimed to confirm the correctness of the algorithm and evaluated the data for the HMM and the CPM. Synthetic data was generated for these simulations and the BF test procedure applied.

The study considered nine different sample sizes (ranging from 50 to 12800) for the simulation. These different sample sizes were coupled with nine different values of the success rate parameter θ in the case of the CPM. Similarly, data was simulated for the HMM, where θ_h was fixed to 0.7, with factorial parameter combinations involving three levels of the switching parameter α and three levels of the success probability d[6], thereby also leading to nine different subplots. Each cell of both simulations was iterated exactly 100 times.

To visualize the results Wetzel et al. used violin plots (Hintze and Nelson, 1998) to represent the distribution of $logBF_{HC}$ coupled with Jeffreys classification scheme. The plots showed the distribution the $logBF_{HC}$ for the previously defined simulations.

[5] At this point the authors introduced integration space restrictions:

"As priors in Bayesian theory can be used to represent theory (Vanpaemel, 2010)(Vanpaemel and Lee, 2012), the uniform integration space is restricted in two ways. First, sticky-state values for α range from 0 to .5, halving the integration space. Second, we eliminate label-switching and consider only those parameter values where θ_h ¿ θ_c, halving the integration space once more—hence, the final integration space covers only a quarter of the unit cube." (Wetzels et al., 2016, p. 202) (For proof I again refer to read the article.)

[6] The success probability d was defined as: $d = \theta_h - \theta_c$(Wetzels et al., 2016, p. 203)

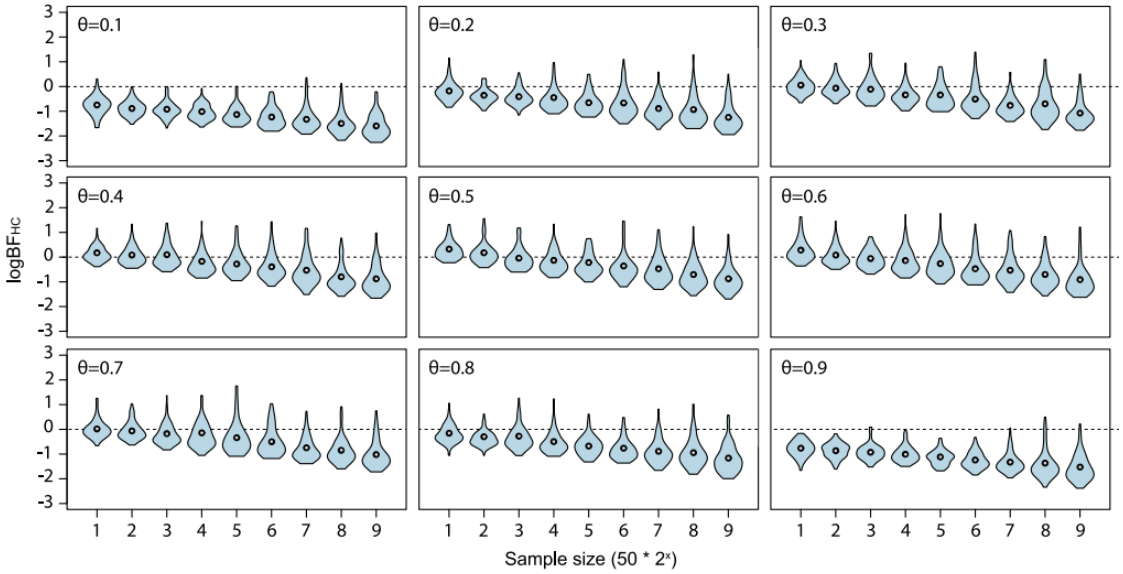


Figure 2: CPM Simulation

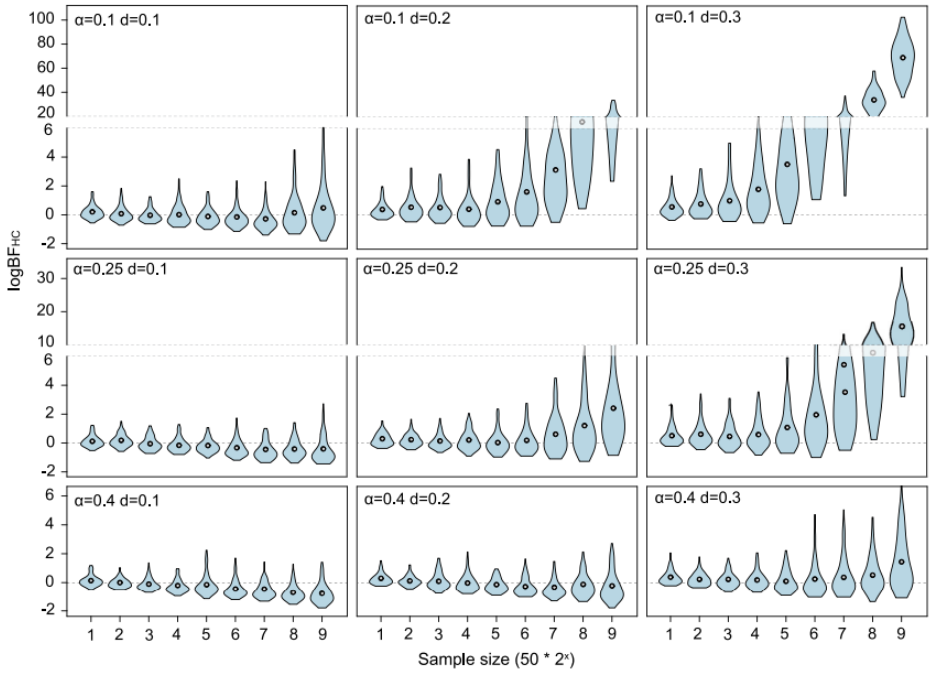


Figure 3: HMM Simulation

The graphs revealed that for small and partly medium sample sizes, the support in favor of the correct classification of the CPM and HMM respectively often failed to exceed the threshold for clear evidence.[7] This suggested that Bernoulli time series were maybe not the ideal fit for evidence in sports streakiness. The authors noted that a surprisingly high BF may arise from model misspecification, and that a BF is more of a measure of relative, not absolute support. In other words when it come to the example application, the data may not be informative enough to distinguish between a streaky and a non-streaky model. The simulation studies as such show that the streakiness is often unimpressive unless the time series is very large or the parameters extreme. (Wetzels et al., 2016, p. 203)

[7] More so with the HMM than the CPM.

2.1.3 Visual Discrimination Task

In addition, the authors presented a second example, beyond the NBA application, to analyse streaky performance. It involved a visual discrimination task where participants were instructed to identify a brightened square on a computer monitor. The brightness was adjusted to achieve three different difficulty levels, which were iterated three times for each participant. The experiment featured four participants, as such yielding 36 time series in total. (Wetzels et al., 2016, p.204f)

Compared were the above introduced BF test alongside a Wald-Wolfowitz runs test (Bradley, 1968) to assess performance patterns. The runs test evaluates the constancy of performance by analysing the number of runs of successes and failures. The BF test compares the evidence for streaky against constant performance. Results indicated that a considerable portion of time series data exhibits streakiness, as determined by both tests. Introduced figures presented a visual representation of the analyzed time series, highlighting streaky and non-streaky patterns. The authors further compared both test results with each other. (Wetzels et al., 2016, p.204f)

The objective was to understand the patterns of human performance in identifying different visual levels. Ultimately, by analyzing the data collected from the participants responses, the experiment aimed to uncover whether their performance exhibited streaky behavior or remained rather constant over time.

2.2 Methodology Extension

As mentioned my goal for the subsequent operationalization is to expand upon the real world example of NBA shooting statistics of the article. My proposal is the following: Narrowing streakiness to only free throws stands in stark disregard to what is actually happening during a basketball game and therefore by extend the streaky performances of players. The test and model by themselves do not pose any issue with this re-proposition but rather the decision to illustrate the BF test solely on the behaviour of free throw shooting.[8] The results of the calculations of such a pre-defined free throw streakiness are not wrong, they just do not fully represent the HHP. I argue this decision misinterprets the actual meaning and understanding of the HHP, as simply too much information is not accounted for by this model design.

With an expansion to overall shooting performances I aim to relay the centre of attention on the game-based flow, or simply said fundamental hot hand streakiness. Here it is important to clarify what type of streak my updated model should focus on. Wetzels et al. observe sample data over a long period of time, six seasons, using the total binary free throw shooting data per game as the base for their streakiness performance assumption. In itself a very logical and reproducible procedure. Choosing to opt for yearly seasonality streakiness is in no way a wrong approach, rather than choosing streakiness of, e.g., a single season or game[9]

Yet, by utilizing such a broadly spanned model, I suggest, something very essential is lost in the model

[8] As suggested by the authors in section 6. on page 204, see:(Wetzels et al., 2016, p.204).

[9] Other very interesting basketball streakiness performances may further include, clutch streakiness (late 4th quarter and overtime (OT)) or single game streakiness, playoff and regular season streakiness or home court and away court streakiness to name the most prominent ones. All these are very fitting for further iterations of the same base BF test to create a comparable overview strategy.

approach. The momentum assumption, or flow, of being *hot* in a current moment, it does not quite fit into such a large time frame. This goes all the way back to defining what a *hot hand*[10] really is. In the case of basketball Scheibehenne describes the phenomenon as: "Many basketball fans believe that players can become "hot", being more likely to score again if they just scored." (Scheibehenne, Wilke, and Todd, 2011, p. 326) I agree with this statement and posit that it is the most widely acknowledged definition. Reasonably during a game a case of a *hot* state would be after, i.e., LeBron James hit three baskets in a row he proceeds to hit one more, then two free throws, and after that two three pointers. This would indubitably be regarded as a *hot* streak.

But if James makes, i.e., two free throws, then proceeds to miss six non free throw shots and after that makes three free throws, but shortly after misses another 4 regular shots again, no one would consider him to be in a hot state – but the model does.[11] Free throws, when it comes to basketball, are too insignificant of a factor to solely attribute an entire players streakiness towards them, let alone their whole game performance. Especially as free throws can be trained for separately and no other player on the court can take any active influence in denying an attempt to shoot it, see: Free Throw Attempt NBA. As such they are not representative of the regular game flow at all.

This I argue is the inherent flaw within the real world NBA example proposition in the article by Wetzels et al., the core streaky development of a players game flow and scoring is simply omitted. All other types of shots[12] are not accounted for, thus the players position, built and streakiness efficiency is reduced to only his binary free throw shooting performance. In the 2005/2006 season for example, Kobe Bryant had his historic 81 point game against the Toronto Raptors, still talked about in the basketball community today. (ESPN, 2005) Everyone who knows about this game would say he was on fire during said game, or was at least in a *hot* state - especially in the 4th quarter, virtually converting every offensive alone. This aspect, I argue, is completely lost in the current model design, signaling the mentioned major loss of information.

I further want to highlight the binary calculation procedure. It works fine in the case of only assessing free throws, as the hit or miss either award 1 or 0 points. Yet there are shots that also award 2 or 3 point as well, if one wants to account for the true underlying *value* of every shot the entire model would have to be readjusted.[13] For the sake of comparison in terms of identical models and steadiness however I will stick with the articles introduced, ceteris paribus, binary assumption. Meaning every hit equalling 1 and every miss equalling 0. It would most definitely be suitable however to expand on a value based procedure in future work to analyse how the streakiness of the analysed players would differ with the implementation of a continuous scale rather than a binary one.

[10] For more context, the first paper discussing the phenomenon was conducted in 1985 (Gilovich, Vallone, and Tversky, 1985).

[11] Importantly, this does not change with the time period or event chosen for the model. In other words the temporal design of the model can be modified.

[12] For different shot examples, see: Non Free Throw Attempt NBA 1-3.

[13] This, of course, suggests that streakiness is influenced by the value of converted shots, not just their frequency. I believe this to be the case, as hitting five free throws in a game in a row is a lot easier than five three pointers. This further makes sense, as streakiness in sports in more of a definition based phenomenon, than a natural one. The mathematical nature behind the streak of hitting five free throws or five three pointers is indifferent but our definition of their respective difficulty, therefore amount necessary to classify as streaky, is highly different.

When creating a model of the HHP it is further imperative to respect for who or what one applies it to. In the case of Shaquille O'Neal and Kobe Bryant for example, totally different player types come to mind.[14],[15] There are five different positions in basketball:

- Point Guard (1 or PG)
- Shooting Guard (2 or SG)
- Small Forward (3 or SF)
- Power Forward (4 or PF)
- Center (5 or C)

O'Neal, for instance, was considered a Center (C), while Bryant mostly took up the role of a Small Forward (SF) or Shooting guard (SG). As such their entire game, from free throw frequency, importance of free throws, their body built[16] to shot selection and endurance was different. I suggest that O'Neal's and Bryant's shot selection inherently have different probabilities of succeeding, simply by the difficulty of their position and subsequent shot attempts. While O'Neal, as seen by the previous comparison by Wetzels et al., is a much more streaky free throw player and less likely to convert a free throw than the more steady Bryant, it is actually the opposite when looking at broader game statistics. O'Neal poses a high 58.2% career field goal percentage (FG%) (ESPN, 2024b) while Bryant had a 44.7% FG%(ESPN, 2024a) over his career.

I consequently posit, as O'Neal took mostly more low-risk conversion field goals, like dunks and short range shots inside the restricted area, compared to Bryant's frequent three point attempts or off-balance jump shots, that he should be not streaky or at least less streaky than Bryant. As such I present this hypothesis statement:

- The null hypothesis (H0): I do not observe any deviation from the free throw results of Wetzels et al.'s model. O'Neal remains classified as a streaky shooter. Bryant remains classified as a non-streaky shooter.
- The alternative hypothesis (H1): I do observe a deviation from the free throw results of Wetzels et al.'s model. O'Neal will be classified as a non-streaky shooter or at least less of a streaky shooter than Bryant. Bryant will be classified as a streaky shooter. Or at least less of a non-streaky shooter than O'Neal.

H1 introduces a slight altercation when compared to H0, my goal here is just to make clear that I expect the streakiness of O'Neal to be less than that of Bryant, not necessarily expecting him to be non-streaky. In mathematical terms:

$$(O'Neal)BF_{HC} < (Bryant)BF_{HC}$$

[14]Which possibly even makes the players unsuitable for comparison. Yet this is a different issue, which will be disregarded.

[15]These two players were analysed with the model by Wetzels et al.

[16]Which starkly affects free throw performance, See: (Cabarkapa et al., 2021).

3. Operationalization

In this section I discuss what data I will make use of and how I classify and process it to ultimately make it suitable for the previously introduced BF test with an underlying HMM and CPM model comparison. I will firstly go about what to expect by scrutinizing the simulation and application data by Wetzels et al. in their article.

3.1 NBA Application

I will now focus on the specific NBA examples utilized in the article and redo the process with the new data and hypothesis of my methodological model expansion.

3.1.1 Wetzels et al. Free Throw Model

For a real world application of the Bayesian test Wetzels et al. chose to analyze binary basketball free throw shooting data from six consecutive NBA seasons. The analysis focused on the free throw shooting performance of two players, Kobe Bryant and Shaquille O'Neal. The results of the BF test revealed consistent support for the CPM over the HMM for Kobe Bryant's free throw shooting across all seasons, indicating non-streaky performance. In contrast, the data consistently favored the HMM over the CPM for Shaquille O'Neal's free throw shooting, indicating streaky performance. (Wetzels et al., 2016, p. 204) So, to put it bluntly, the exact opposite of what I proposed in my hypothesis statement.

The authors correctly illustrated the behavior of the test for each player's performance, providing log BF values in favor of the CPM over the HMM for Kobe Bryant and in favor of the HMM over the CPM for Shaquille O'Neal. The extent of the support ranged from BF_{CH}[17] $= \frac{1}{exp(-0.26)} = 1.3$ in 2008 to $BF_{CH} = \frac{1}{exp(-0.86)} = 2.6$ in 2006 for Kobe Bryant, and from $BF_{HC} = exp(0.20) = 1.2$ in 2010 to $BF_{HC} = exp(6.99) = 1085.7$ in 2006 for Shaquille O'Neal.[18] (Wetzels et al., 2016, p. 204)

The visual representation of the results through the log BF values provides a clear indication of the support for streaky and non-streaky performance in basketball free-throw shooting. The consistent favoring of the CPM over the HMM for Kobe Bryant and the HMM over the CPM for Shaquille O'Neal's performance is visually evident in the graphical representation of the log Bayes factor values for each season. It further becomes clear that Kobe Bryant shoots at a much better rate overall, visible by p(success), than Shaquille O'Neal. This may indicate that more successful free throw shooters are much more consistent in their game than others.[19] (Wetzels et al., 2016, p. 205)

[17] For the necessary fracture calculation of the negative exp. function the BF definition was switched to CH to achive the desired output.

[18] A reminder for the results here is that a negative log indicates evidence in favour of the CPM, a positive log favour for the HMM.

[19] The nature of free throws is highly intriguing in itself, as I mentioned they are trained for separately and basically represent an entire new game when compared to normal basketball circumstances.

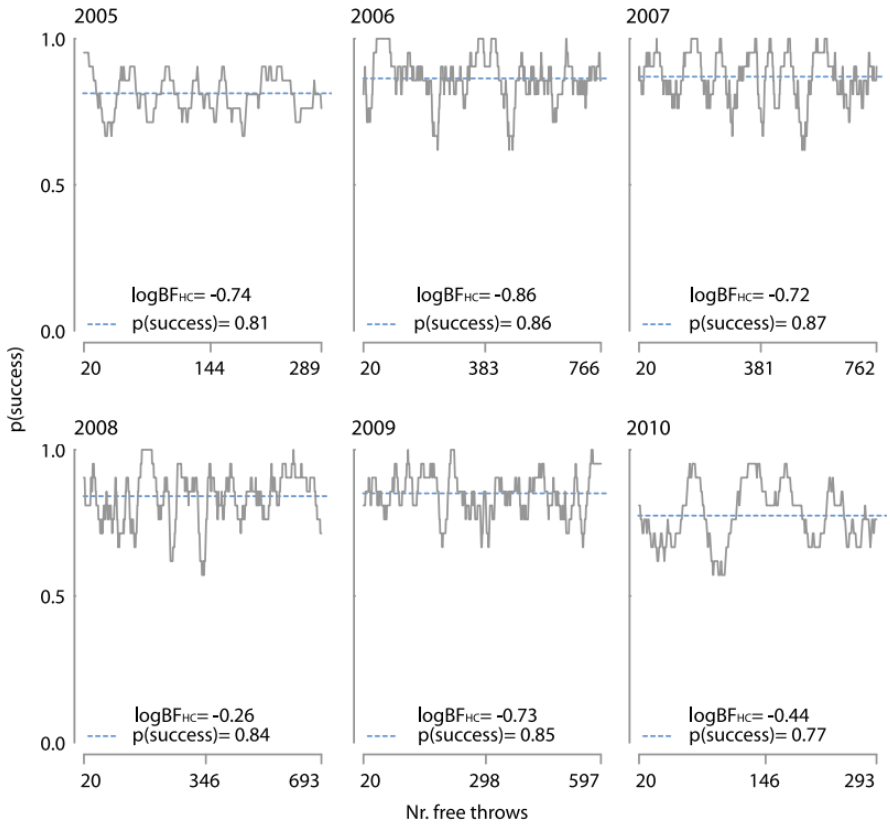


Figure 4: Kobe Bryant Free Throw Streakiness

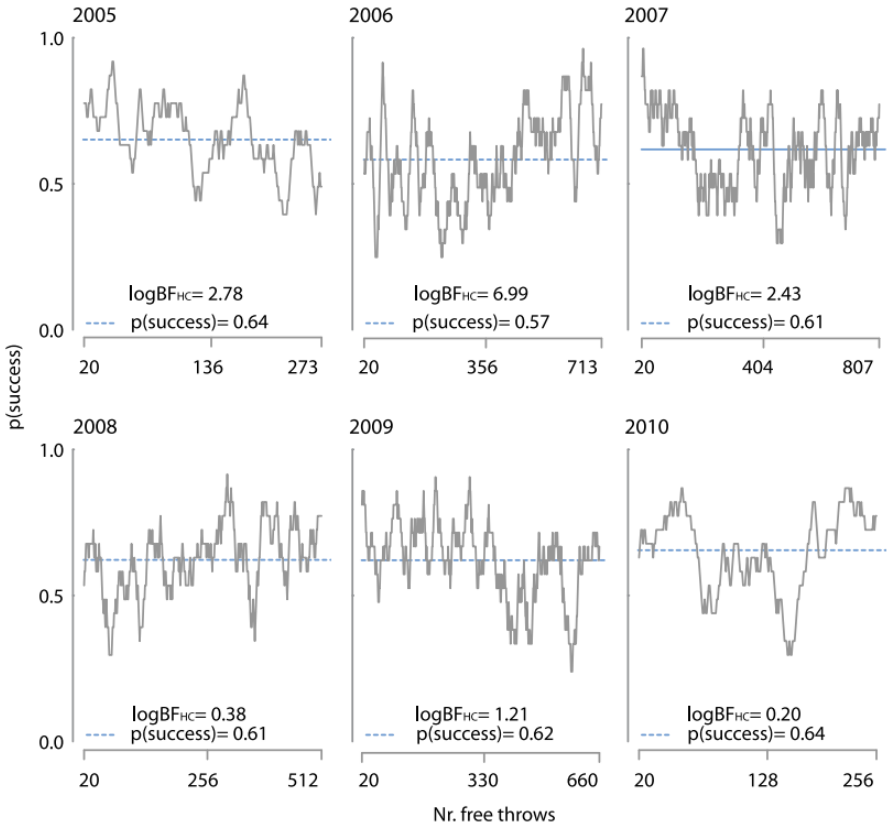


Figure 5: Shaquille O'Neal Free Throw Streakiness

3.1.2 Model Expansion

As mentioned, the core model process will not change at all but will simply be reapplied to different data. For straightforward results I chose to iterate the model code over the first three full games of both Shaquille O'Neal and Kobe Bryant in the 2005/2006 NBA season and their best and worst performance game[20] of that season.

I will use coherent shooting data of both players in these games recorded on ESPN of the 2005/2006 NBA season. The calculation of the BF test with respect to the previously defined HMM and CPM classifications will be done separately in R.[21] Naturally, as this only applies to data for ten games, a considerable expansion may pose very different results and could most definitely be explored, leading to a different outcome. Yet, the goal of the relatively small number of games is to not lose track of the goal of reflecting on the game-based HHP definition of this paper.

The model resulted in streaky behaviour/HMM support for Shaquille O'Neal in 4 out of 5 cases, all of them were very small however and classified as not more than a bare mention. In 1 out of the 5 analysed games O'Neal's results classified non-streaky behaviour/support for the CPM. For Kobe Bryant the model was continuously in favour of the HMM or streaky behaviour in 5 out of 5 instances in respect to Jeffreys classification scheme, albeit also all classifying as not worth more than a bare mention. As such H0 can be rejected and H1 in turn accepted.[22]

3.2 Results

I will now compare the expected results of the model[23] to the actual result and those of Wetzels et al.

In their example, the authors applied the Bayesian test to analyze streaky performance in basketball free-throw shooting. The results showed varying degrees of evidence for streakiness, with Kobe Bryant's free throw shooting data consistently favoring the CPM over the HMM, while Shaquille O'Neal's data consistently favored the HMM over the CPM. The authors highlighted the difficulty in finding compelling evidence for or against streakiness and emphasize that the support in favor of the hot hand in sports is mixed, as evidenced by previous studies. (Wetzels et al., 2016, p. 206)

I proposed the actual streakiness of these players to be different based on their position, play style and subsequent overall shooting performances rather than their free throw performances. My results showed that this assumption holds, although not very strongly. This may be due to the scale of data being compared, Wetzels et al. applied greater binary data samples than I did.[24] Consequently the streakiness of the games may not have been sensitive and sufficient enough to classify in Jeffreys classification scheme. Despite 9 out of 10 samples classifying in the same level, Bryant's 5 HMM scores were on average higher

[20] Measured simply in terms of highest and lowest points scored in a single game during that season.

[21] For insight into the code I created a GitHub repository, see: Replication Data.

[22] To be mentioned here is that the evidence is not strong at all, another set of 20 games may change the outcome - nonetheless the proposal holds.

[23] Reminder, I expected H0 to not hold.

[24] They used entire seasons, around 300-700 instances per iteration. Meanwhile I used data per game for one iteration, the maximum here was 66 the lowest 12. Further, a reminder to (2.1.2 Simulation Study) where the simulation showed that smaller samples have a harder time identifying streakiness clearly.

than O'Neal's 4 HMM scores.[25] In other words, even without the negative classification in O'Neal's case his streaky performances are still less streaky than those by Bryant, which is coherent with my hypothesis assumption.

In conclusion, while this Bayesian test provides valuable insights into streaky performance, the mixed evidence and inherent ambiguity in discriminating streaky performance call for a critical reevaluation of the test's effectiveness in capturing the dynamics of streakiness in different performance domains. Further research and exploration of alternative models or methodologies are essential to advance the understanding of streaky performance.

4. Discussion and Conclusion

Overall, the article by Wetzels et al. provides a significant contribution to the understanding of streaky performance in sports and psychology. It offers a rigorous statistical approach to assess the HHP, it highlights the challenges in discriminating an HMM from a CPM, particularly in Bernoulli time series, and further suggests that the evidence for streakiness in sports may be influenced by the informativeness of the data. Moreover, its approach to investigate a players streakiness performance on only a certain kind of performance, like free throws, is not faulty, but when evaluating for the HHP one has to keep its core definition in mind. A hot hand is mostly a very fleeting moment, lasting maybe a quarter of a game, especially the last, where a player puts his entire team on his back to achieve a win.[26]

The authors themselves proposed several options for expansion. The inclusion of subject-specific knowledge in the prior distributions for the parameters for instance, or exploring the possibility of collecting and analysing continuous variables instead of binary variables, which I also briefly touched upon in my methodology extension. The authors further proposed extending the competing models by allowing the switching probability to depend on the state or adding a hierarchical structure that simultaneously takes into account multiple seasons and multiple players.[27] (Wetzels et al., 2016, p. 206)

Thus further extensions could include contextual factors on streaky performance. For instance, the impact of game pressure, player fatigue, or opponent defensive strategies in basketball. Another factor, besides the position aspect, which was not accounted for by the authors was the career aspect of both Bryant and O'Neal during the observed seasons. O'Neal (1992) started his professional NBA career four years earlier than Bryant (1996). Bryant (2016) also retired 5 years later than O'Neal (2011). So when Bryant won his 4th and 5th championships in 2009 and 2010 respectively, O'Neal was nearing his retirement. Therefore the selected seasons and roles of players could also be adjusted for.

Moreover, the concept extends beyond sports, suggesting that random positive feedback can reinforce specific beliefs and behaviors in various contexts. And while I did not go into detail about the visual discrimination task, Wetzels et al. correctly identified, that such a phenomenon has a lot, maybe even

[25] Bryant's average score: 1.31166644 in comparison to O'Neal's cleaned HMM average (excluding the negative CPM score): 1.2125555

[26] A classic example of a late hot hand, the clutch, in U.S. American sports culture.

[27] Albeit these being a logical extensions I only focused on redrawing the article's NBA application model. Analysing all would have diluted a clear goal for this paper.

everything, to do with psychology and self confidence in such a moment. If a player truly believes to hit the next basket and has been in such a situation before, there may indeed be a slim statistical improvement in hitting the basket, like a placebo effect. While this is not at all proven further work in the field of self identification tasks may pose to be vital to fully grasp the picture of the HHP which is still widely known as the hot hand fallacy. Maybe that fallacy association will change with time.

Appendix

Photography



Figure 6: Free Throw Attempt NBA



Figure 7: Non Free Throw Attempt NBA 1



Figure 8: Non Free Throw Attempt NBA 2



Figure 9: Non Free Throw Attempt NBA 3

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